

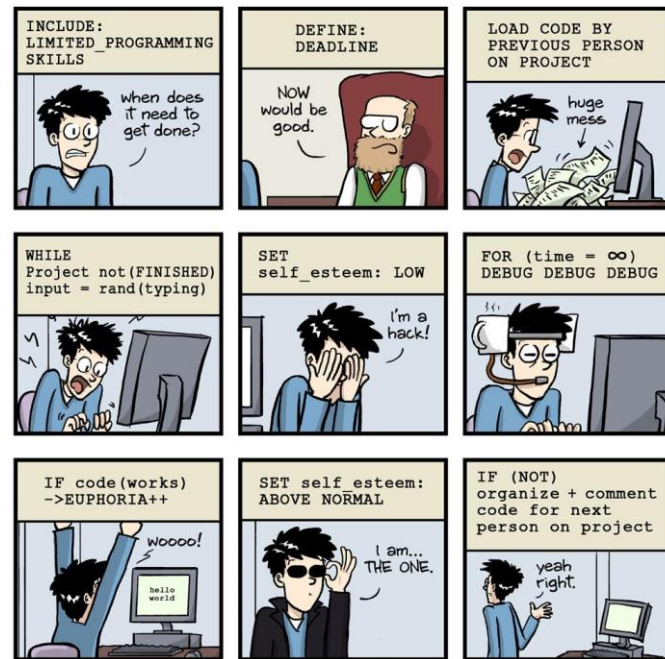


NTNU – Trondheim
Norwegian University of
Science and Technology

Introduction to Python

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PROGRAMMING FOR NON-PROGRAMMERS



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Outline

- This hour:
 - What is Python
 - How to use Python
 - Tips and tricks, packages
 - Example -> Function and dictionary
- Next hour:
 - Using optimization tools in Python -> Pyomo
 - Example: Door manufacturer
 - Danger: Side effects may include preferring Python over other programming languages! User discretion is advised

Learning outcome and goal

- Basic knowledge on Python
- Learn how to code in Python and use software like Spyder
- Learn about practical packages
- Learn programming tricks
- Learn to create optimization problems in Python
- **Note: Some basic programming language experiences are expected here.**
 - **Supplementing with guides on Youtube can be helpful**

What is Python

- Up and coming programming language, created in 1991
- Open-source programming language
- Popularity increasing
- Regularly updated
- **Recently seen as the more common and go-to language at NTNU**



Monty Python comedy group

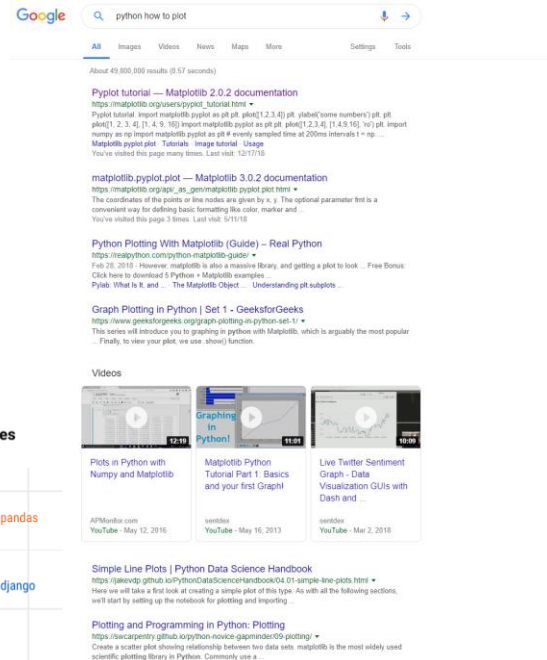
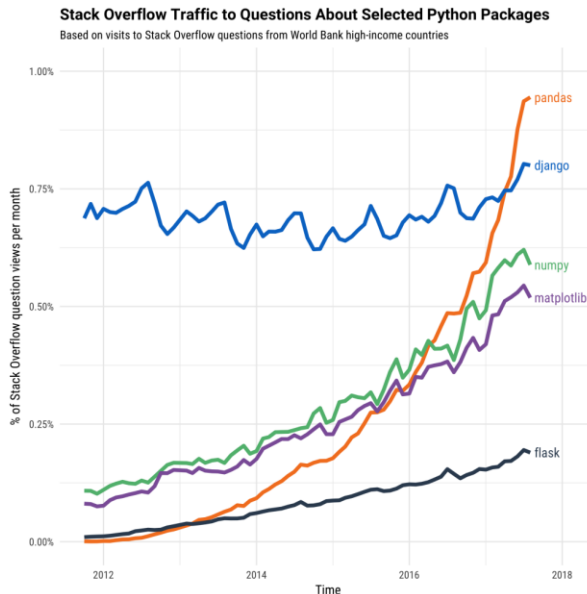
TIOBE predicts "It is very likely that Python will enter the top 3 and even might become the new number 1 in the long run".

1	1		Java	16.881%	+3.92%
2	2		C	14.966%	+8.49%
3	3		C++	7.471%	+1.92%
4	5	▲	Python	6.992%	+3.30%
5	6	▲	Visual Basic .NET	4.762%	+2.19%
6	4	▼	C#	3.541%	-0.65%
7	7		PHP	2.825%	+0.63%
8	8		JavaScript	2.411%	+0.31%
9	-	▲	SQL	2.316%	+2.32%
10	14	▲	Assembly language	1.409%	-0.40%
11	11		Swift	1.384%	-0.44%
12	12		Delphi/Object Pascal	1.372%	-0.45%
13	17	▲	MATLAB	1.366%	-0.25%
14	18	▲	Objective-C	1.358%	-0.15%
15	10	▼	Ruby	1.182%	-0.78%
16	9	▼	Perl	1.175%	-0.82%
17	16	▼	Go	0.996%	-0.65%
18	15	▼	R	0.965%	-0.80%
19	13	▼	Visual Basic	0.922%	-0.89%
20	21	▲	PL/SQL	0.702%	-0.51%

The latest TIOBE index.
Image: TIOBE

What is Python

- People can develop packages to expand the possibilities
 - The packages are accessible online
- Much help online for troubleshooting
 - Google Search
 - Stackoverflow
- Good community



How to install Python

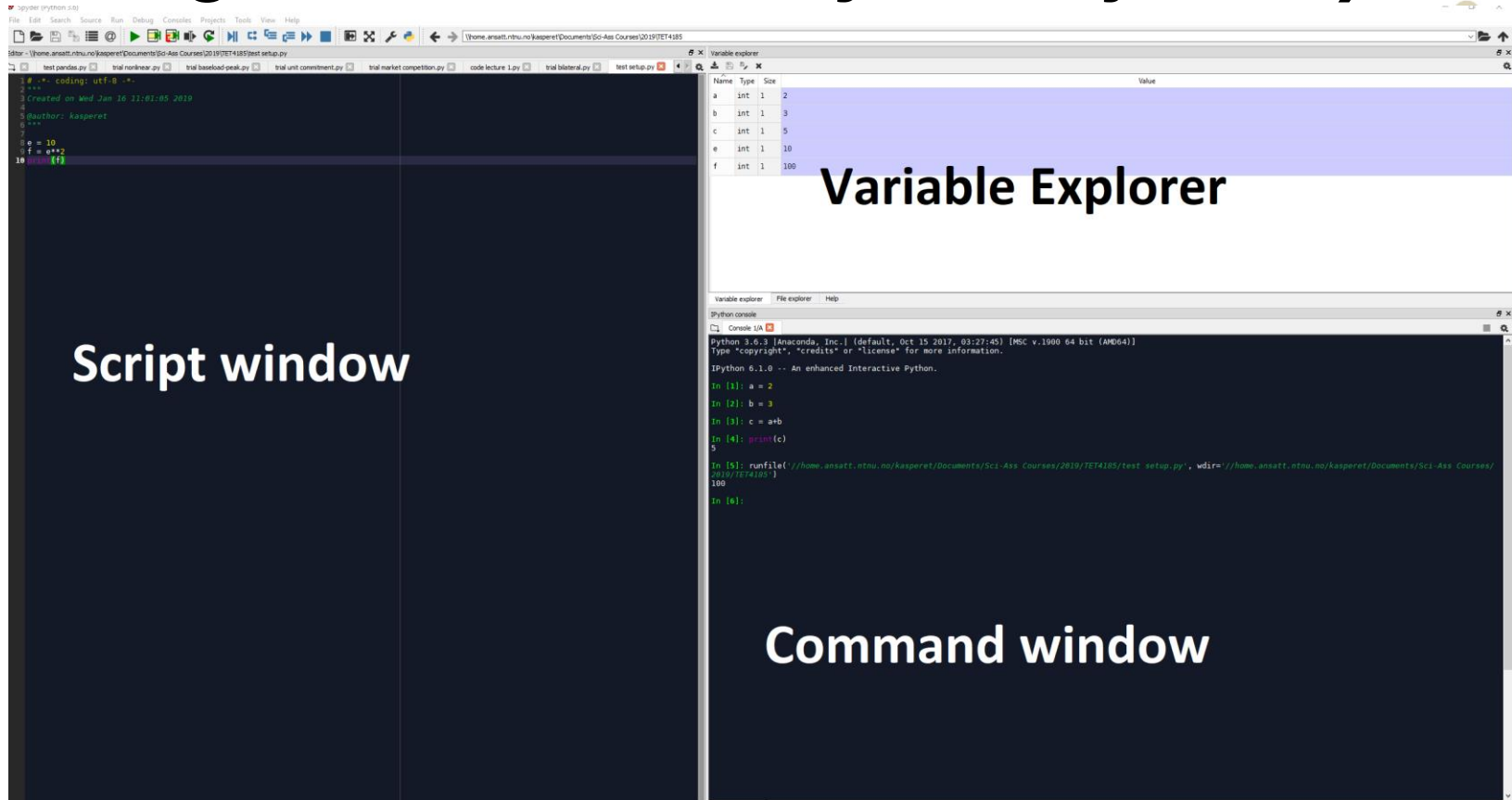
- Please see own presentation about this!
 - Python Setup Guide.pptx
 - Available at Blackboard
- This presentation uses Anaconda for python-related setup
- Other distributions are possible for use of Python:
<https://code.visualstudio.com/docs/languages/python>



Python via Anaconda

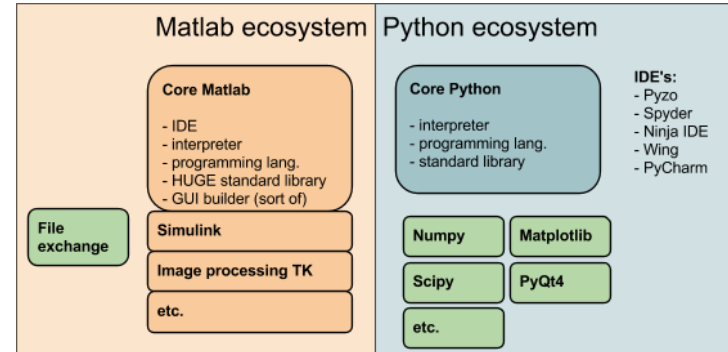
- Anaconda:
 - “**Anaconda** is a free and open-source distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment.” **Wikipedia**
- Many packages included, and easy access to install further packages.
- Can run python through «Spyder» (IDE):
 - Shows command window, console window, variable explorer and more
 - Pretty much same layout as MATLAB (own option actually to have it equal to MATLAB layout)
- **Note:** You're free to use your own Python-setup, this is the one I started with myself
 - Other examples include: Visual Studio code (VS Code – Very practical when getting used to)

Spyder Layout (note that I use a dark background, can easily be adjusted)



Specifics about Python

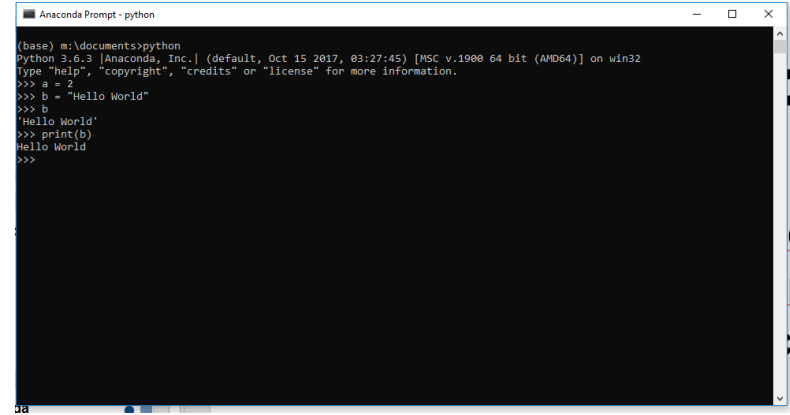
- No ; at the end of a line needed
- Indent-sensitive
- Has no specific «integrated development environment» (IDE) (Spyder is a 3rd party IDE)
- Need to install packages manually to extend possibilities
- NB: Index starts at 0!
- Use # for commenting



```
1 # -*- coding: utf-8 -*-
2 """
3 Created on Wed Jan 16 11:01:05 2019
4
5 @author: kasperet
6 """
7
8 a = 2
9 b = "Hello World"
10
11 print(b)
12
13
14 a = 10
15 for i in range(0,a):
16     print(i)
17
18 |
```

How to use Python

- You can either start coding in command window/console window or through scripts
- Python starts indexing at 0!
- Note the indent after the for-loop (line 16)



```

Anaconda Prompt - python

(base) m:\documents>python
Python 3.6.3 [Anaconda, Inc.] (default, Oct 15 2017, 03:27:45) [MSC v.1900 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>> a = 2
>>> b = "Hello World"
>>> b
'Hello World'
>>> print(b)
Hello World
>>>

```

```

1 #-*- coding: utf-8 -*-
2 """
3 Created on Wed Jan 16 11:01:05 2019
4
5 @author: kasperet
6 """
7
8 a = 2
9 b = "Hello World"
10
11 print(b)
12
13
14 a = 10
15 for i in range(0,a):
16     print(i)
17
18 |

```



```

In [2]: runfile('//home.ansatt.n
2019/TET4185')
Hello World
0
1
2
3
4
5
6
7
8
9

```

Using packages

- Python has a lot of packages to simplify problems
- Problem: Create array of 10x1
 - Time consuming if big arrays to manually add each value
 - Solution: numpy package
 - Numpy is a great mathematical package with lots of helpful functions
 - Our case: `array_list = numpy.zeros((100000))`
 - We can create the array size right away, and put the wanted values when we have them
 - Own experience: Sped up a 5 minute process down to 20 seconds (array of approx 8760x33x12)

```
7
8 import numpy as np
9 import sys
10 import time
11 import pandas as pd
12 import pyomo.environ as pyo
13 from pyomo.opt import SolverFactory
14
```

Tips when getting code to run

- Check what packages are imported at the beginning
- Read the comments during the coding and try to understand what is going on
(few comments -> irritating...)
- If unsure, proceed to download each package to ensure you have them
- This is usually only necessary when you just started programming in Python

My personal list of frequently
used packages

```
7  
8 import numpy as np  
9 import sys  
10 import time  
11 import pandas as pd  
12 import pyomo.environ as pyo  
13 from pyomo.opt import SolverFactory  
14
```

Tips and tricks

- Numpy is a very practical package
- When storing much different data, dictionaries are helpful
- Pandas is a great package for data handling, although a bit difficult to learn right away
- Practice makes perfect, try and solve a problem, and search if stuck. One can only get better

```
46 dic = {}
47 dic['Name'] = "Kasper"
48 dic['name'] = "Kasperuss"
49 dic['Age'] = 24
50 dic['Attributes'] = {'Skills':5, 'Creativity':6, 'Without ball':1}
51 dic['Team'] = "NTNU BIL"
52
53 print("Player name: " + dic['name'])
54 print("Player age: " + str(dic['Age']))
55 for attribute in dic['Attributes']:
56     print(attribute + " -> " + str(dic['Attributes'][attribute]))
57
```

```
9
Player name: Kasperuss
Player age: 24
Attribute: Skills -> 5
Attribute: Creativity -> 6
Attribute: Without ball -> 1
```

dic - Dictionary (5 elements)

Key	Type	Size	Value
Age	int	1	24
Attributes	dict	3	{'Skills':5, 'Creativity':6, 'Without ball':1}
Name	str	1	Kasper
Team	str	1	NTNU BIL
name	str	1	Kasperuss

pandas

$$y_{it} = \beta' x_{it} + \mu_i + \epsilon_{it}$$



Tips and tricks (2)

- Functions are helpful to specify objective of code
- Necessary for certain actions (optimization)
- Makes the coding much more cleaner, sorted and easier to understand
- Be sure to comment alot, necessary to understand and remember what is going on

```
58 def FindBestAttribute(Data):  
59     """  
60     Function that finds the best attribute for the player, and prints it  
61     """  
62     #Attribute name  
63     attr = ""  
64     #Attribute value  
65     value = 0  
66     #For every attribute  
67     for attribute in Data["Attributes"]:  
68         #If the attribute value for the current attribute is higher than the current stored one  
69         if value <= Data["Attributes"][attribute]:  
70             #Change attribute name to the current one  
71             attr = attribute  
72             #Change attribute value to the current one  
73             value = Data["Attributes"][attribute]  
74         print("Best attribute: " + attr + " -> " + str(value))  
75     return  
76 #Run function  
77 FindBestAttribute(dic)
```

Best attribute: Creativity -> 6

Questions?

